Reflection structure against M, $\phi_* = 5\,M_P$

$\beta = 1.2$ | $\beta = 2$ | $\beta = 2.9$

kinetic fraction $\pi^2/6$

parks

ρ_χ/ρ_R (dissipation)

vs lab.

T_J [GeV]

$\log_{10}\mathcal{R}$, first impact

elastic: no particle production

M/M_P

$M = 4 \times 10^{-31}$ (laboratory) M_P, dissipation (rows 1, 2)
$M = 10^{-15}\,M_P$, dissipation (rows 1, 2)
$M = 10^{-12}\,M_P$, dissipation (rows 1, 2)
$M = 10^{-3}\,M_P$, natural mode ($\alpha = 0$)
$M = 0.5\,M_P$, natural mode ($\alpha = 0$)
$\alpha = 0.001$ (row 2)
$\alpha = 0.01$ (row 2)
$\alpha = 0.08$ (row 2; row 1 in blue)
$\alpha = 0.15$ (row 2)
$M = 10^{-9}\,M_P$, $\alpha = 0.08$ (row 2, strip)
$M = 10^{-6}\,M_P$, $\alpha = 0.08$ (row 2, strip)
$M = 10^{-3}\,M_P$, $\alpha = 0.08$ (row 2, strip)
$M = 0.5\,M_P$, $\alpha = 0.08$ (row 2, strip)
BBN, 2 MeV – 20 keV (indicative)
$\mathcal{R}$ at the first impact, main store (row 3)
$\mathcal{R}$ at the first impact, large-M diagnostic (row 3)
briefing notes' $M_{\rm crit}$, median ($1.2 \times 10^{-15}\,M_P$)
briefing notes' $M_{\rm crit}$, maximum ($8.2 \times 10^{-13}\,M_P$)